Heterogeneous 3D SSR Integration-Point Study, Relative-Residual Variant

April 8, 2026

1 Goal

This note studies how tetrahedral integration-point count changes the continuation response of the standard heterogeneous 3D SSR benchmark on `SSR_hetero_ada_L1`. This residual-stop variant repeats the same `P1`, `P2`, and `P4` element comparison as the delta-λ study, but switches the continuation Newton stop back to the old default `relative_residual = 1e-4`. The comparison focuses on continuation-curve shape, runtime, final load level at a shared horizon, and numerical stability.

2 Benchmark Setup

The shared comparison target is $\omega_{\mathrm{final}} = 6.7000 \times 10^6$. Every run uses the same continuation settings as the companion delta-λ report except for the continuation Newton stop: `secant` indirect continuation, `step_max = 100`, `d_lambda_diff_scaled_min = 1e-4`, and Newton `relative_residual = 1e-4`. The initial Newton solve also uses `relative_residual = 1e-4`. The execution environment is fixed at `mpirun -n 8` with `OMP_NUM_THREADS = 1`, `node_ordering = block_metis`, `constitutive_mode = overlap`, and the PETSc-side `PETSC_MATLAB_DFGMRES_HYPRE_NULLSPACE` linear solver.

The published matrix is element-specific: `P1` uses `q1`, `q4`, `q11`, `q24`, and `q45`; `P2` uses `q4`, `q11`, `q24`, and `q45`; and `P4` is reported only for `q24` and `q45`. This keeps the `q11` rule as the established `P2` baseline in the code base while dropping the unhelpful `P4/q11` failure from the published comparison. The `q45` curve is treated as the within-element reference because it is the richest rule in the published comparison matrix.

One methodological clarification matters here: the standard `SSR_hetero_ada_L1` benchmark reaches the familiar $\omega \approx 6.7 \times 10^6$ scale only on the indirect continuation path. The earlier direct-study omega scale was therefore not the same problem, so this corrected study uses the standard indirect heterogeneous 3D SSR capture path throughout. The wall-clock policy is also element-specific: `P1` and `P2` retain a 1.2000×10^3 s guard, while `P4` is allowed to run without a wall-clock timeout so the higher-order curves can finish unless a real watchdog abort is triggered by solver pathologies such as repeated zero-correction stalls.

3 Reference-Element Quadrature Rules

The tetrahedral rules used in this study are defined on the unit reference tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. Table 1 summarizes the five published rules. The point coordinates and weights are taken directly from the implementation in `src/slope_stability/fem/quadrature.py`, and the point clouds are visualized in Figure 1. Marker size is proportional to $|w|$ and red markers indicate negative quadrature weights.

All five rules integrate over the same reference volume, so the weights sum to $1/6$ in every case. The low-order `q1` and `q4` rules use only positive weights; `q11` introduces one negative-weight point and is the established `P2` baseline in this repository; `q24` is the first fully positive high-order rule in the published set; and `q45` again contains one negative-weight point while supplying the richest spatial sampling used in the comparison.

Table 1: Reference-tetrahedron quadrature-rule summary for the published comparison set.

Rule	Points	Negative weights	$w_{\min}$	$w_{\max}$	Used in study for
q1	1	0	0.166667	0.166667	P1
q4	4	0	0.0416667	0.0416667	P1/P2
q11	11	1	-0.0131556	0.0248889	P1/P2
q24	24	0	0.00167954	0.0092262	P1/P2/P4
q45	45	1	-0.039327	0.0138301	P1/P2/P4

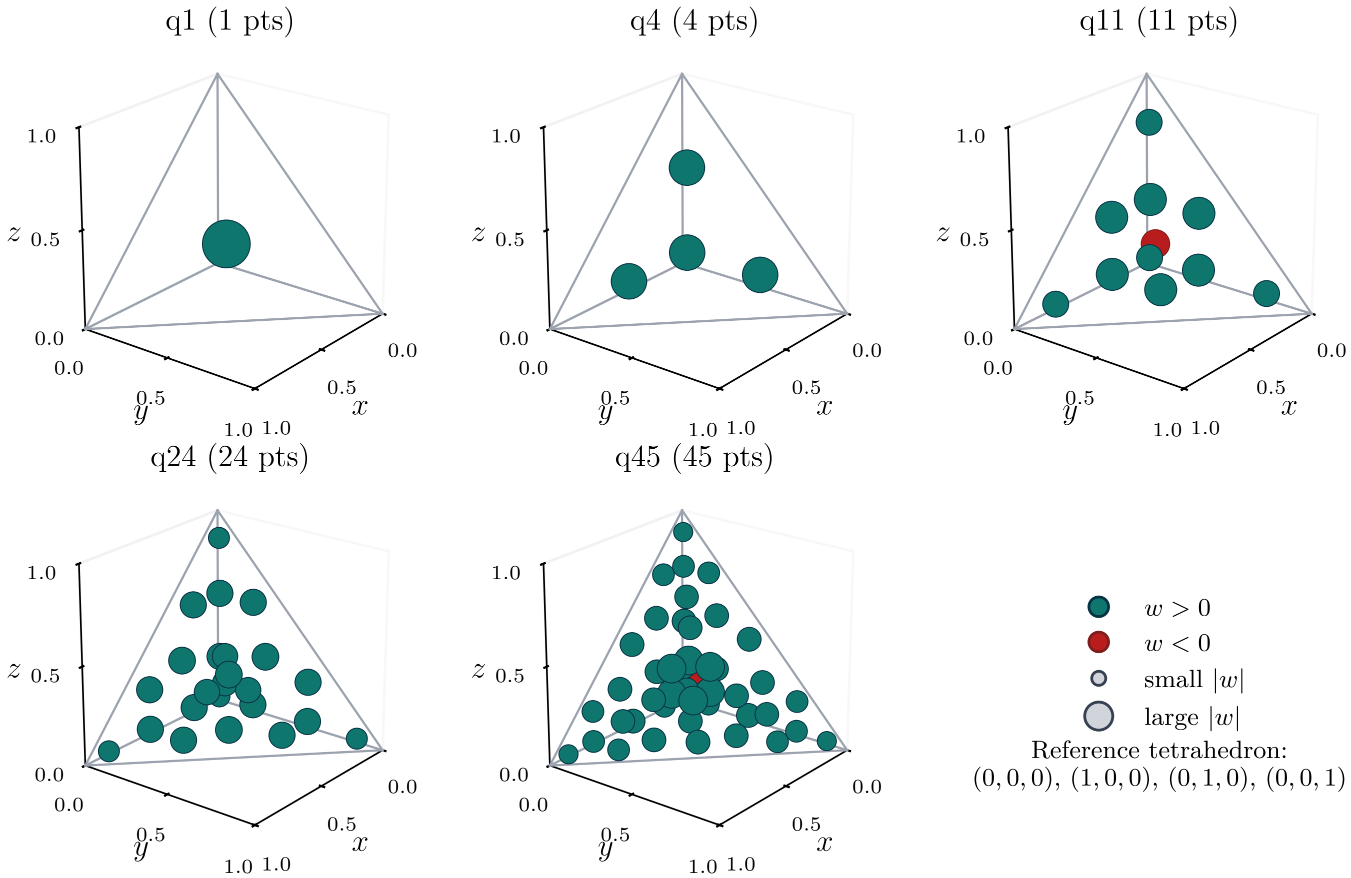

Figure 1: Reference-tetrahedron quadrature points for `q1`, `q4`, `q11`, `q24`, and `q45`. Marker size scales with $|w|$, and negative-weight points are shown in red.

4 Continuation Curves

The continuation curves are the primary comparison target. Figure 2 groups the runs by element family, with each column showing the integration-point comparison inside one element type. This is the main visual comparison view for the study.

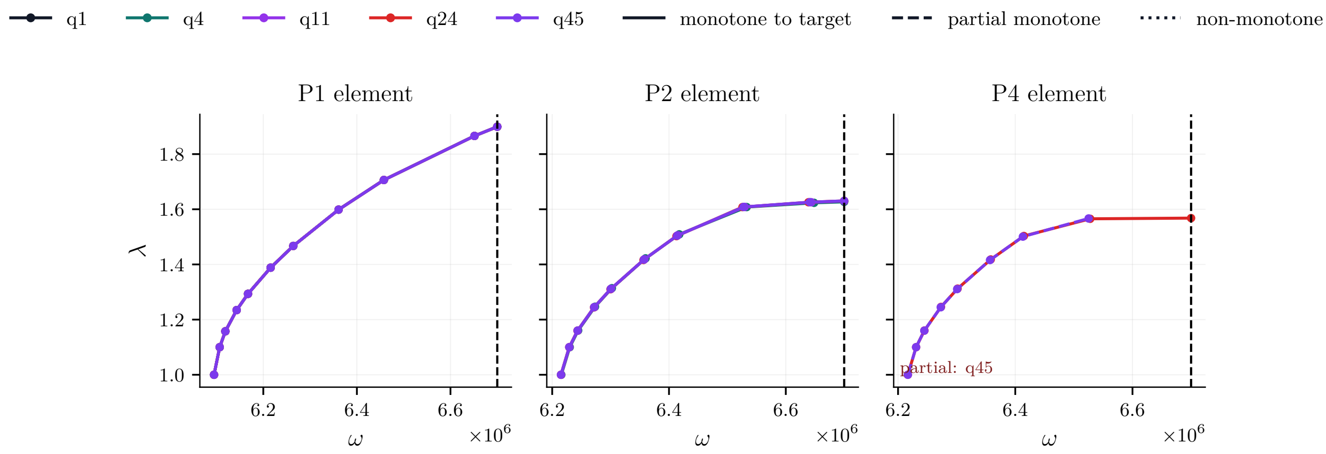

Figure 2: Continuation curves grouped by element family. The dashed vertical line marks the shared target ω_{final}. Solid curves reached the shared target monotonically, while dashed curves denote monotone partial branches if a watchdog abort occurred before the target. Rules with no monotone branch are listed inside the affected panel.

5 Timings And Target Results

The timing and target-value summaries below use the same element-first layout as the continuation plot. Hatched bars indicate runs that did not reach the shared target monotonically; the timing bar still reports their measured wall time, while the target-value plot suppresses their $\lambda(\omega_{\text{final}})$ bar because no like-for-like interpolation is available.

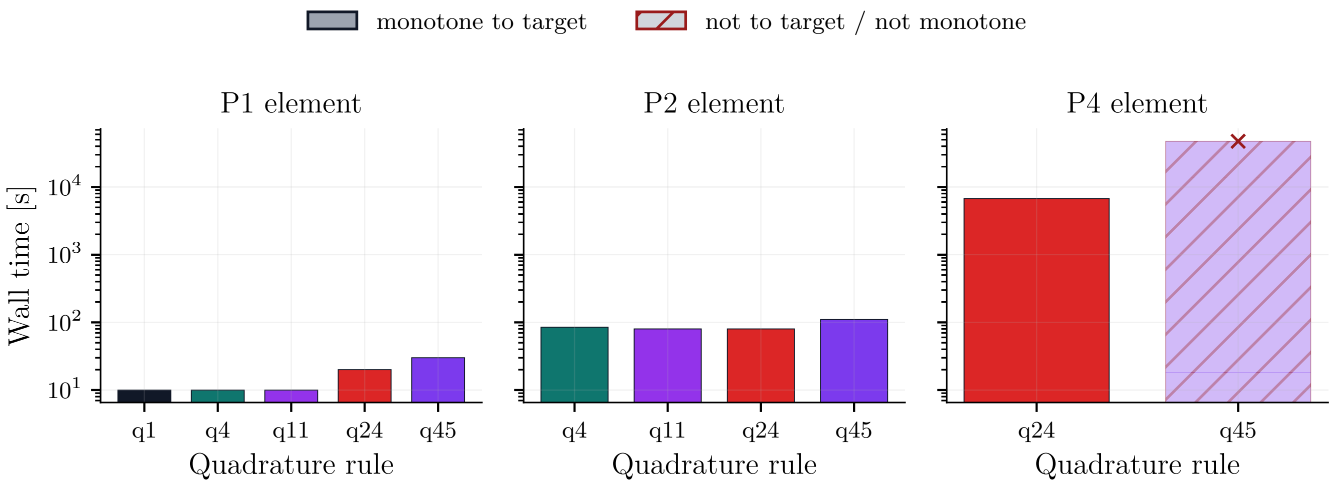

Figure 3: Wall time by quadrature rule, organized by element family.

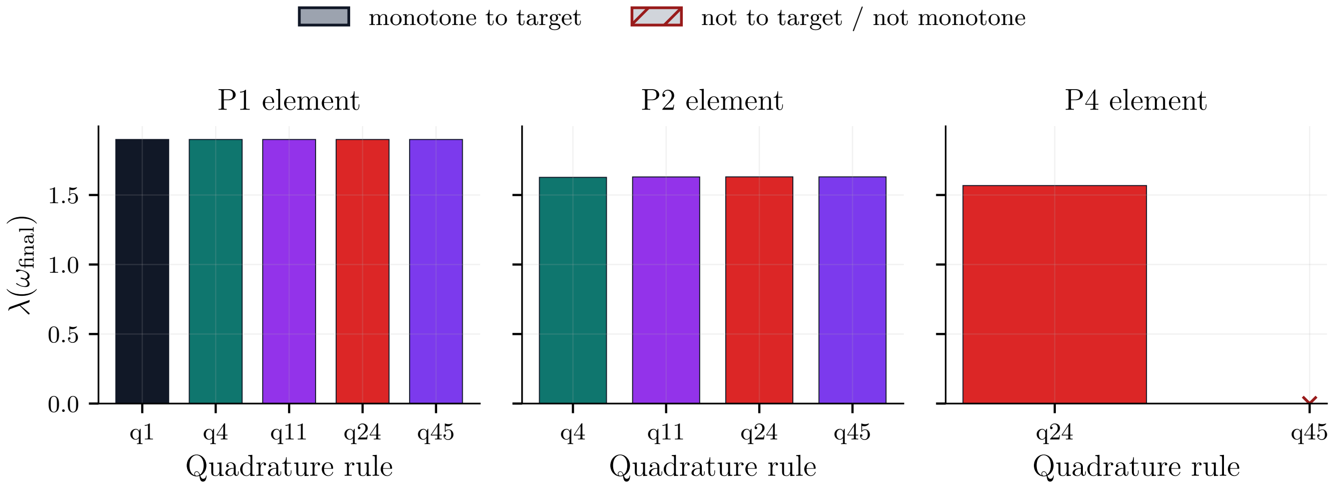

Figure 4: Interpolated $\lambda(\omega_{\text{final}})$ for target-reaching monotone runs, organized by element family.

Table 2: Main study results. "Monotone to target" requires success, target reach, and non-decreasing ω and λ histories.

Element	Rule	Time [s]	Steps	$\lambda(\omega_{\text{final}})$	$\Delta\lambda_{\text{vs }q45}$	ω_{last}	Stop reason	Monotone to target
P1	q1	10	11	1.89927	-1.572e-13	6.7e+06	omega max stop	yes
P1	q4	10	11	1.89927	-2.28e-13	6.7e+06	omega max stop	yes
P1	q11	10	11	1.89927	8.948e-14	6.7e+06	omega max stop	yes
P1	q24	20	11	1.89927	-1.068e-13	6.7e+06	omega max stop	yes
P1	q45	30	11	1.89927	0	6.7e+06	omega max stop	yes
P2	q4	85	10	1.627	-0.003452	6.7e+06	omega max stop	yes
P2	q11	80	10	1.62997	-0.0004881	6.7e+06	omega max stop	yes
P2	q24	80	10	1.63039	-6.323e-05	6.7e+06	omega max stop	yes
P2	q45	110	10	1.63046	0	6.7e+06	omega max stop	yes
P4	q24	6734	9	1.56797	–	6.7e+06	omega max stop	yes
P4	q45	47391	8	–	–	6.52514e+06	last-step maxit	no

6 Findings

Out of 11 element/quadrature combinations, 11 produced a monotone continuation branch and 10 reached $\omega_{\text{final}} = 6.7000 \times 10^6$ monotonically. The report treats the highest quadrature rule, $q45$, as the within-element reference curve.

For P1, the fastest monotone curve was q11 at 10.0000 s. Relative to the q45 reference, the largest absolute shift in $\lambda(\omega_{\text{final}})$ came from q4 with $|\Delta\lambda| = 228.0000 \times 10^{-15}$.

For P2, the fastest monotone curve was q24 at 80.0000 s. Relative to the q45 reference, the largest absolute shift in $\lambda(\omega_{\text{final}})$ came from q4 with $|\Delta\lambda| = 3.4520 \times 10^{-3}$.

For P4, the fastest monotone curve was q24 at 6.7340×10^3 s. The furthest monotone branch under the study settings was q24 at $\omega_{\text{last}} = 6.7000 \times 10^6$. Partial monotone branches: q45.

7 Nonfinished Runs

The `P4/q45` run produced a monotone accepted branch through step 8, reaching $\omega_{\text{last}} = 6.5251 \times 10^6$. The next continuation solve targeting $\omega = 6.7000 \times 10^6$ exhausted the Newton cap of 200 iterations with terminal relative residual 5.8095×10^{-3}. The continuation code then started a reduced-step retry, but that retry was intentionally terminated and is not promoted into the published branch. The report therefore freezes the accepted branch at the previous point and classifies the missing final step as a last-step maxit failure.

8 Reading The Tables

The table reports the wall time for each run, accepted continuation steps, the terminal ω reached, the interpolated $\lambda(\omega_{\text{final}})$ when that interpolation is valid, and the shift in $\lambda(\omega_{\text{final}})$ relative to the `q45` curve inside the same element family. This makes it possible to separate three effects:

- continuation-curve agreement inside one element family,
- cost growth as integration-point count increases,
- and outright instability or inability to reach the shared horizon.

A Reference Quadrature Point Tables

Reference rule q1

Table 3: Reference tetrahedron points and weights for q1. Coordinates are reported on the unit tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

Pt	x	y	z	$1-x-y-z$	w
1	0.25	0.25	0.25	0.25	0.16666667

Reference rule q4

Table 4: Reference tetrahedron points and weights for q4. Coordinates are reported on the unit tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

Pt	x	y	z	$1-x-y-z$	w
1	0.1381966	0.1381966	0.1381966	0.5854102	0.041666667
2	0.5854102	0.1381966	0.1381966	0.1381966	0.041666667
3	0.1381966	0.5854102	0.1381966	0.1381966	0.041666667
4	0.1381966	0.1381966	0.5854102	0.1381966	0.041666667

Reference rule q11

Table 5: Reference tetrahedron points and weights for q11. Coordinates are reported on the unit tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

Pt	x	y	z	$1-x-y-z$	w
1	0.25	0.25	0.25	0.25	-0.013155556
2	0.071428571	0.071428571	0.071428571	0.78571429	0.0076222222
3	0.78571429	0.071428571	0.071428571	0.071428571	0.0076222222
4	0.071428571	0.78571429	0.071428571	0.071428571	0.0076222222
5	0.071428571	0.071428571	0.78571429	0.071428571	0.0076222222
6	0.39940358	0.10059642	0.10059642	0.39940358	0.024888889
7	0.10059642	0.39940358	0.10059642	0.39940358	0.024888889
8	0.10059642	0.10059642	0.39940358	0.39940358	0.024888889
9	0.39940358	0.39940358	0.10059642	0.10059642	0.024888889
10	0.39940358	0.10059642	0.39940358	0.10059642	0.024888889
11	0.10059642	0.39940358	0.39940358	0.10059642	0.024888889

Reference rule q24

Table 6: Reference tetrahedron points and weights for q24. Coordinates are reported on the unit tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

Pt	x	y	z	$1-x-y-z$	w
1	0.35619139	0.21460287	0.21460287	0.21460287	0.0066537917
2	0.21460287	0.21460287	0.21460287	0.35619139	0.0066537917
3	0.21460287	0.21460287	0.35619139	0.21460287	0.0066537917
4	0.21460287	0.35619139	0.21460287	0.21460287	0.0066537917
5	0.87797812	0.040673959	0.040673959	0.040673959	0.0016795352
6	0.040673959	0.040673959	0.040673959	0.87797812	0.0016795352
7	0.040673959	0.040673959	0.87797812	0.040673959	0.0016795352
8	0.040673959	0.87797812	0.040673959	0.040673959	0.0016795352
9	0.03298633	0.32233789	0.32233789	0.32233789	0.0092261969
10	0.32233789	0.32233789	0.32233789	0.03298633	0.0092261969
11	0.32233789	0.32233789	0.03298633	0.32233789	0.0092261969
12	0.32233789	0.03298633	0.32233789	0.32233789	0.0092261969
13	0.26967233	0.063661002	0.063661002	0.60300566	0.0080357143
14	0.063661002	0.26967233	0.063661002	0.60300566	0.0080357143
15	0.063661002	0.063661002	0.26967233	0.60300566	0.0080357143
16	0.60300566	0.063661002	0.063661002	0.26967233	0.0080357143
17	0.063661002	0.60300566	0.063661002	0.26967233	0.0080357143
18	0.063661002	0.063661002	0.60300566	0.26967233	0.0080357143
19	0.063661002	0.26967233	0.60300566	0.063661002	0.0080357143
20	0.26967233	0.60300566	0.063661002	0.063661002	0.0080357143
21	0.60300566	0.063661002	0.26967233	0.063661002	0.0080357143
22	0.063661002	0.60300566	0.26967233	0.063661002	0.0080357143
23	0.26967233	0.063661002	0.60300566	0.063661002	0.0080357143
24	0.60300566	0.26967233	0.063661002	0.063661002	0.0080357143

Reference rule q45

Table 7: Reference tetrahedron points and weights for q45. Coordinates are reported on the unit tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

Pt	x	y	z	$1-x-y-z$	w
1	0.25	0.25	0.25	0.25	-0.039327007
2	0.61758719	0.12747094	0.12747094	0.12747094	0.0040813161
3	0.12747094	0.12747094	0.12747094	0.61758719	0.0040813161
4	0.12747094	0.12747094	0.61758719	0.12747094	0.0040813161
5	0.12747094	0.61758719	0.12747094	0.12747094	0.0040813161
6	0.90376351	0.03207883	0.03207883	0.03207883	0.00065808677
7	0.03207883	0.03207883	0.03207883	0.90376351	0.00065808677
8	0.03207883	0.03207883	0.90376351	0.03207883	0.00065808677
9	0.03207883	0.90376351	0.03207883	0.03207883	0.00065808677
10	0.4502229	0.049777096	0.049777096	0.4502229	0.0043842588
11	0.049777096	0.4502229	0.049777096	0.4502229	0.0043842588
12	0.049777096	0.049777096	0.4502229	0.4502229	0.0043842588
13	0.049777096	0.4502229	0.4502229	0.049777096	0.0043842588
14	0.4502229	0.049777096	0.4502229	0.049777096	0.0043842588
15	0.4502229	0.4502229	0.049777096	0.049777096	0.0043842588
16	0.31626955	0.18373045	0.18373045	0.31626955	0.013830064
17	0.18373045	0.31626955	0.18373045	0.31626955	0.013830064
18	0.18373045	0.18373045	0.31626955	0.31626955	0.013830064
19	0.18373045	0.31626955	0.31626955	0.18373045	0.013830064
20	0.31626955	0.18373045	0.31626955	0.18373045	0.013830064
21	0.31626955	0.31626955	0.18373045	0.18373045	0.013830064
22	0.022917788	0.23190109	0.23190109	0.51328003	0.0042404374
23	0.23190109	0.022917788	0.23190109	0.51328003	0.0042404374
24	0.23190109	0.23190109	0.022917788	0.51328003	0.0042404374
25	0.51328003	0.23190109	0.23190109	0.022917788	0.0042404374
26	0.23190109	0.51328003	0.23190109	0.022917788	0.0042404374
27	0.23190109	0.23190109	0.51328003	0.022917788	0.0042404374
28	0.23190109	0.022917788	0.51328003	0.23190109	0.0042404374
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32	0.022917788	0.23190109	0.51328003	0.23190109	0.0042404374
33	0.51328003	0.022917788	0.23190109	0.23190109	0.0042404374
34	0.73031343	0.037970048	0.037970048	0.19374648	0.0022387397
35	0.037970048	0.73031343	0.037970048	0.19374648	0.0022387397
36	0.037970048	0.037970048	0.73031343	0.19374648	0.0022387397
37	0.19374648	0.037970048	0.037970048	0.73031343	0.0022387397
38	0.037970048	0.19374648	0.037970048	0.73031343	0.0022387397
39	0.037970048	0.037970048	0.19374648	0.73031343	0.0022387397
40	0.037970048	0.73031343	0.19374648	0.037970048	0.0022387397
41	0.73031343	0.19374648	0.037970048	0.037970048	0.0022387397
42	0.19374648	0.037970048	0.73031343	0.037970048	0.0022387397
43	0.037970048	0.19374648	0.73031343	0.037970048	0.0022387397
44	0.73031343	0.037970048	0.19374648	0.037970048	0.0022387397
45	0.19374648	0.73031343	0.037970048	0.037970048	0.0022387397